Supplementary Materials for:
"Learnable Role-Conditioned Evidence Sentence Selection with Interpretable Mixture Reranking"

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This supplement documents FEVER conversion details, training hyperparameters, and additional split statistics for the main paper. All quantitative tables use human-gold FEVER evidence labels.

S1 FEVER Conversion

We convert the filtered FEVER evidence-selection release into the C^{2}GES sentence-ID schema. For each claim–document instance:

- Wikipedia lines become sentences with IDs $s000, s001, \ldots$;
- human evidence line IDs become gold evidence sentence IDs;
- the claim becomes the query;
- SUPPORTS/REFUTES becomes the role label.

Instances without recoverable gold evidence lines are discarded. Conversion scripts are in `source/code/prepare_fever_benchmark.py`.

S2 Training Hyperparameters

Table S1: Training and model settings

Setting	Value
Encoder	sentence-transformers/all-MiniLM-L6-v2 (frozen)
Role head	MLP on $[h_s; h_q; e_r]$, hidden size 256
Mixture floors	$(0.35, 0.25, 0.05)$ then renormalize
Optimizer	Adam, learning rate 1×10^{-3}
Epochs	4 (checkpoint by development F1)
Train / Dev / Test	4000 / 800 / 800
K	3
Loss	pairwise mixture loss + 0.5× role-head pairwise loss
Bootstrap resamples	400 document-cluster draws

S3 Development Split Summary

Table S2: Development-split evidence F1 ($K = 3$)

Method	F1
Lead-K	0.4644
LexCue	0.4563
TF-IDF	0.4951
SBERT	0.4894
BM25	0.5039
Query-only	0.5028
No-role	0.5006
No-graph	0.5077
Full C^{2}GES	0.5081

S4 Selected Mixture Weights

The best development checkpoint uses mixture weights approximately $(0.6992, 0.2507, 0.0502)$ for (w_q, w_r, w_g).

S5 Artifact Map

- Benchmark JSON: `workspace/fever_benchmark/{train,dev,test}/`
- Manifest: `workspace/fever_benchmark/manifest.json`
- Run summary: `workspace/fever_runs/learnable_role/summary.json`
- Checkpoint: `workspace/fever_runs/learnable_role/checkpoint.pt`
- Training code: `source/code/c2ges_learnable.py`